Tatul Danielyan

AI Researcher at Metric AI Lab • Yerevan, Armenia

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Website | GitHub | LinkedIn | Google Scholar

Research Interests

Robot learning and physical AI: reliable robot execution, failure detection and recovery, robot evaluation, and embodied AI.

Education

National Polytechnic University of Armenia Sep 2023 – Jun 2027 (expected)
B.S. in Artificial Intelligence

Experience

AI Researcher Jun 2024 – Present
Metric AI Lab *Yerevan, Armenia*

- Built evaluation pipelines for robot manipulation, including automated success and failure detection [1].
- Fine-tuned and deployed vision-language-action (VLA) policies for manipulation tasks.
- Designed teleoperation pipelines and curated demonstration datasets for robot policy training.
- Developed simulation environments for policy training and evaluation.
- Built LLM and RAG systems for production deployment [2].

Publications

[1] Z. Navasardyan, **T. Danielyan**, H. Davtyan. *FailBench: How Reliable are VLMs at Judging Robot Task Success?* arXiv:2609.03611, 2026. [project]

[2] **T. Danielyan**, M. Avetisyan, H. Davtyan. *Cloud and On-Premises Deployment of Uzbek Legal RAG via Targeted Retriever Fine-Tuning.* EMNLP 2026 Industry Track (poster), arXiv:2608.29284. [project]

Talks

- *Retrieval for legal QA in low-resource languages* [2], Jina AI Birds-of-a-Feather, EMNLP Oct 2026

Awards

- 2nd Place (team), DO2025 Drug Discovery ML Challenge 2025
- 2nd Place (team), WaterData Hackathon 2024

Skills

Robot learning & ML: PyTorch, LeRobot, Hugging Face Transformers, OpenCV; imitation learning (diffusion policy, ACT)
Robotics & simulation: ROS, ROS 2, MuJoCo, Isaac Sim, Isaac Lab, Genesis
Engineering: Python, Linux, CUDA, Docker, Git, SLURM, Weights & Biases; distributed (multi-GPU) training
Spoken languages: Armenian (native), English, Russian